

TankMatic™ R1

Installation and Operation Manual

7 Sensor Non-Intrusive Water Level Monitor and Pump Dry-Run Protector
with Remote Access



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List of Abbreviations

- IOM – Installation and Operation Manual.
- IP - Internet Protocol.
- MQTT - Message Queuing Telemetry Transport.
- QR - Quick Response.
- SPin - SuperMation Products identification number.
- SPL - SuperMation Products Limited.
- UI - User Interface.
- URL - Uniform Resource Locator or website link.

Models Applicable:

- R1/L1
- Pump 'Dry-Run' protection relay.

1 Prerequisites

Safety is crucial so please ensure proper safety rules and use of tools are followed. The details of these are outside the scope of this document. SPL is not liable for any injuries or damages incurred due to incorrect installation.

1.1 Tools

Before installation, ensure the following tools are available:

1. Hot glue gun (minimum 60Watt) for full size (11mm / 0.43inch) sticks.
2. Extension cord (length depends on the location of the target tank and nearest power receptacle).
3. Ladder or scaffolding for tanks installed at significant height.
4. Clean, lint-free, cloth and rubbing alcohol in a spray bottle.

1.2 Site resources

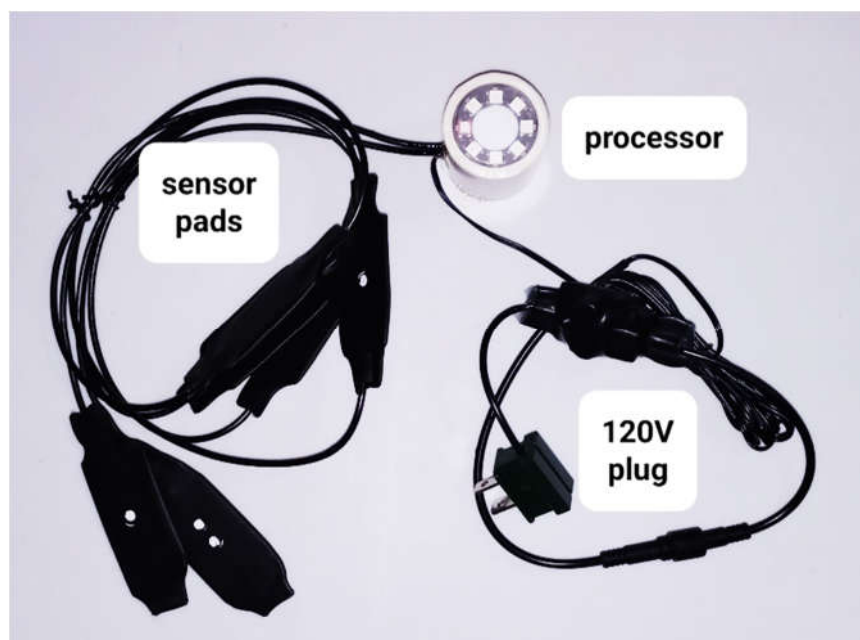
The following are needed for the installation at the tank:

1. One available 120VAC NEMA power receptacle at the target tank(s).
2. WiFi 2.4GHz network (signal strength should be -70dBm or stronger). Internet for remote access and notifications.

1.3 Kit contents

The kit contains all that is needed for a full installation as listed below:

1. One TankMatic hub and external sensor pads.
2. One 120V power cord.
3. One hot glue stick 11mm diameter.



2 Connect the product to the WiFi network

2.1 Power up

Insert the 2 prong power plug into a 120VAC receptacle, preferably within 20 feet of the WiFi router.

2.2 Enter WiFi credentials

The mode LED on the Visualizer (see [subsection 3.2.1](#)) will flash blue for approximately 30 seconds then flash cyan for 120 seconds. This pattern will repeat until it successfully connects to a router. The cyan flashes indicate it is in configuration mode. Use your phone to connect to its private network:

SSID: TM1 (the number may be different if multiple products are installed).

Password: 1234567890

When connected use the UI URL in the receipt email or scan the below QR code (this works only when connected to the R model private network):



The UI will load, tap on the **About** tab then scroll down to the WiFi credentials section:

TankMatic™ >...

Model: 7S-120-R1K

Normal mode: Mrj +++.. (-57dBm)

WiFi credentials

Scan WiFi

Saved SSID | Password: Mrj | [REDACTED]

Step 1A tap the relevant SSID:

1: Mrj | -57

2: ARRIS-D638 | -61

3: CarMatic | -86

4: Amplia-CC1D | -86

5:

Or step 1B for hidden SSID: **Save SSID**

Step 2 Password: **Save password**

There are 5 buttons showing the discovered SSIDS followed by their signal strength in dBm (the strongest signal is at button 1). Tap the button of the relevant SSID or enter your hidden SSID manually, tap the **Save SSID** button, then tap the back button on your phone. Enter the relevant password then

tap the **Save password** button. The product will restart into normal mode. The mode LED will flash blue several times. If the credentials are correct and the router accepts the product, the light will turn yellow or green.

If the product did not connect successfully, repeat this [sub-section \(3.2\)](#) and carefully re-enter the credentials. If the saved SSID and password are correct and connection is not successful, please check the WiFi router settings.

2.3 Remote Access

2.3.1 Telegram App

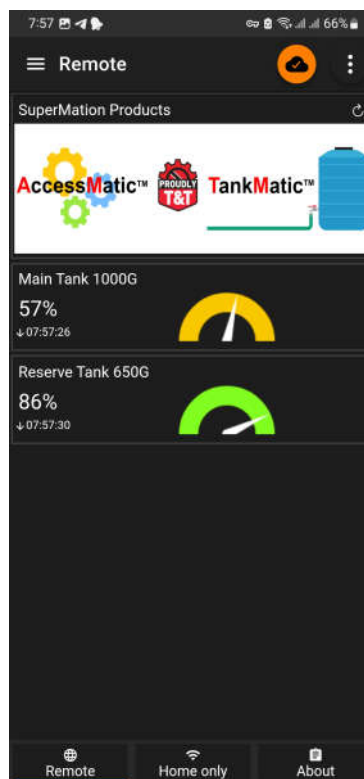
If provided, use the provided URL in the receipt email to add the product in Telegram. Tap the **Start** button and the keyboard menu will now be visible. Do not share this information with unauthorized individuals. The product must be used at least once every 6 months or else the profile is removed by the Telegram network. The Sales Agreement does not cover reprogramming of the product.

2.3.1.1 Add or remover Users

Tap the **Status** button, to add the first user type **AU1** or remove the first user **RU1**. Correspondingly for the second user use **AU2** or **RU2**.

2.3.2 IoT Central App

The login details will be provided upon purchase of the product. The level is displayed in real-time on the app dashboard as shown in the example below.



3 Installation onto the Tank

The product is designed for plastic (polypropylene) water storage tanks. Ceramic, concrete, metal and silicone tank walls are not compatible with the product. It is strongly recommended to install onto a full tank. The tank outer surface must be dry with no trace of moisture. Please do not install in drizzle/rain.

3.1 Powering

The 120VAC plug needs to be inserted into a nearby receptacle as in the example below.



Hot glue can be used to secure the wire to the tank surface, pipes and concrete walls for example. The LEDs will go active. On a full tank, Visualizer ring should have 7 green LEDs (the 8th or bottom LED is the mode indicator). The mode LED will flash blue a couple times then turn solid blue for connection to your router or extender. At this time you should get a message from Telegram if you were added as a user for notifications. You should receive notifications in the Telegram app as the sensors read the water level. Tapping the **Status** button will give the estimated water level. On the IoT Central app, the tank level will be displayed. If the tank is partially full, at least one LED on the Visualizer will be red from a clockwise direction.

3.2 Install the components

Please wear silicone gloves or finger protectors. If hot glue gets on the skin, do not rip it off! Spray rubbing alcohol onto the glue and it will de-bond for easy removal.

All components should be placed equally apart. When all locations are identified, clean thoroughly with rubbing alcohol.

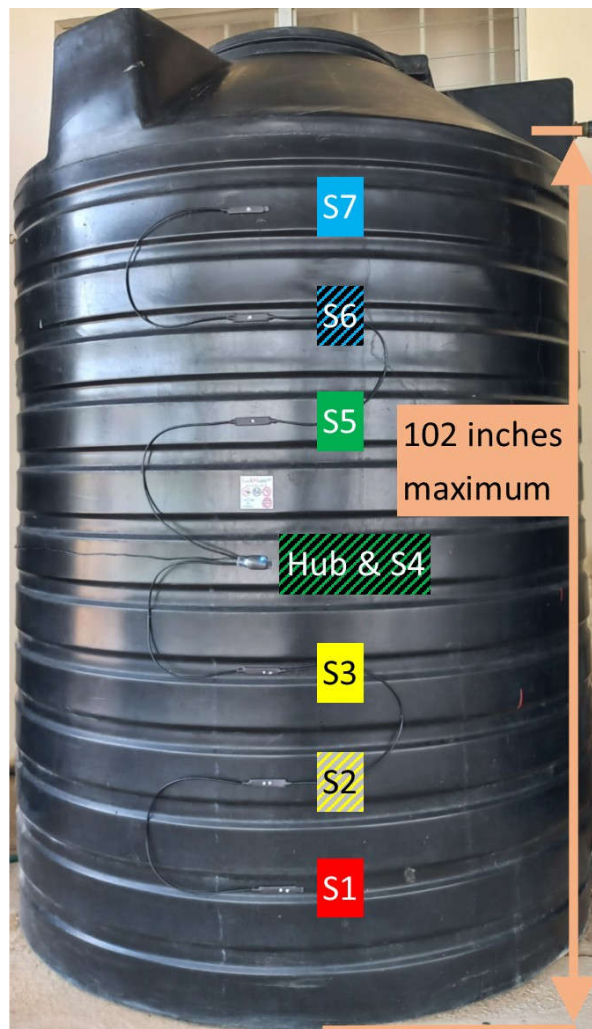
For tanks significantly taller than 102 inches from fill port to base, additional Nodes are recommended.

Ensure your hot glue gun is hot enough that some glue dribbles from the nozzle. Starting with the Hub, apply a rectangle of glue at the sensor area (behind the LED). Ensure the cables are oriented downward or to the side (never upwards). Firmly press the hub onto the cleaned tank surface and hold for 20 seconds. Each sensor pad requires a line of glue on the side opposite the dot marker. The upper sensors has one dot and goes above the Hub, the lower sensors has two dots and goes below.



If the installation was incorrectly done, spray the cooled glue with rubbing alcohol and let soak for 2 minutes. Carefully pry off the component (do not pull the cable!) with a plastic scraper and peel off traces of the glue. Repeat the installation properly.

If purchased, apply the same technique to the Node(s) and its external sensor pads.

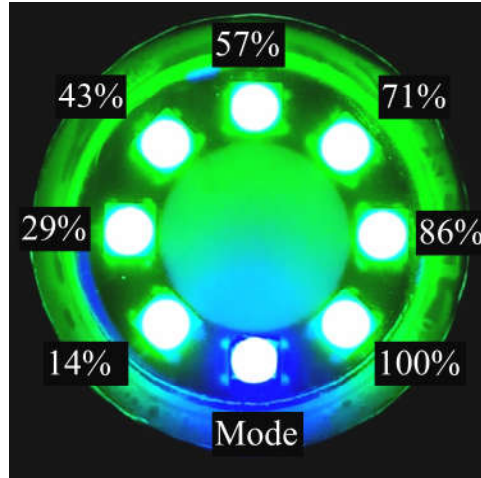


The above shows a typical installation.

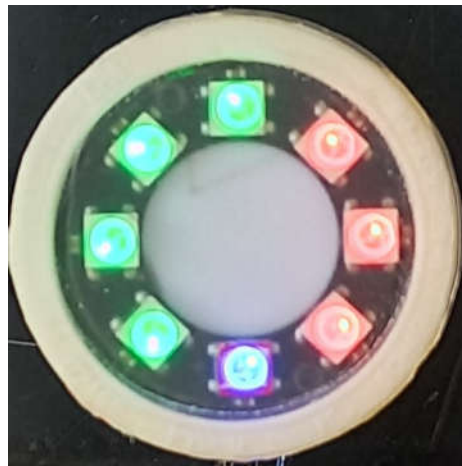
3.2.1 Visualizer Ring

3.2.1.1 Level Monitor (R model)

The visualizer is a LED string that allows for easy reading of the water level. A full tank shows all green (except the mode LED). As the water drops, the LEDs will change to red in a clockwise direction. An empty tank will be shown as all red.



The above picture shows a full tank. The below tank shows a tank approximately 57% full.

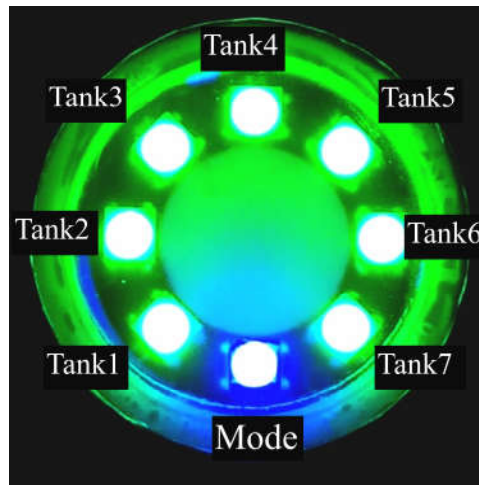


The mode LED is summarized in the table below.

Colour	Mode
Blue	Normal mode/WiFi Connected
Flashing Blue	Reconnect to WiFi
White	Calibration/Updates
Cyan	Configuration mode
Purple	Standalone mode

3.2.1.2 Pump Protector (L model)

When the product is configured as a pump protector, up to 7 tanks can be monitored simultaneously. The cable length of each sensor is limited to 10ft so the spacing and geometry of the tanks is important for the installation.



Each tank has its status. Green for water above the sensor and red for water below the sensor. If any of the tanks are critical in a parallel bank of tanks, the pump will be stopped. When the water rises above all the sensors, the pump will be enabled after a 2 minute safety delay.

The mode LED remains the same as the R model in [sub-section 3.2.1.1](#).

3.2.2 Pump Dry-Run Protection Relay

The protection relay prevents dry-run damage to the pump. The relay plugs into the power outlet and the pump plugs into the relay. When the level goes critical, the pump will be stopped. When the level rises above the lowest sensor pad, the pump will be enabled after approximately 2 minutes (the pump running depends on its pressure sensor/switch).



The relay has an LED to indicate its state. Red is pump off. Green is pump enabled. Purple means reconnecting to the level monitor. The UI has controls for the relay in the Control tab.

TankMatic™ ...>

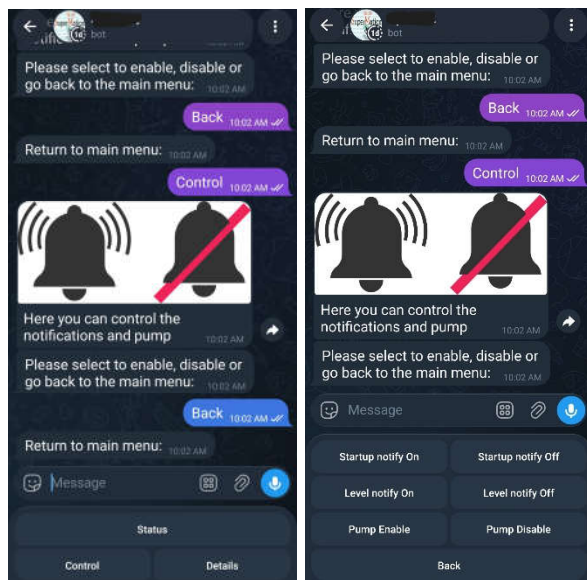
Model: 7S-120-R1K

Normal mode: Mrj +++.. (-56dBm)

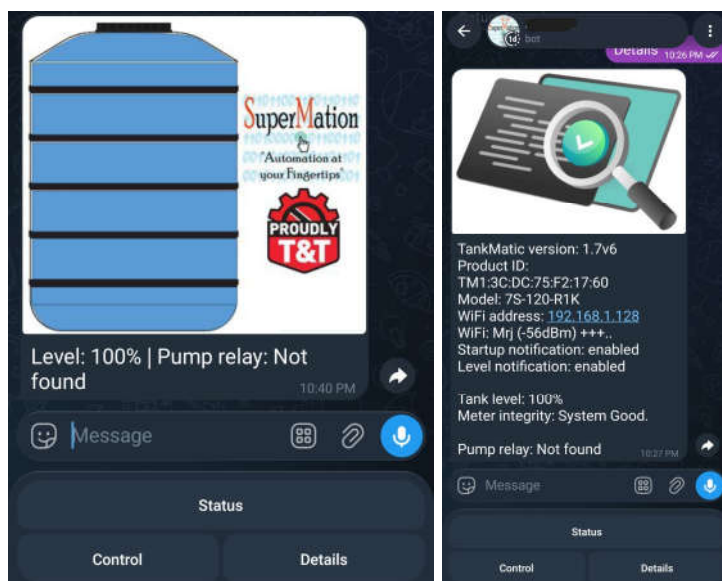
Pump Control

Pump Enable Pump Disabled

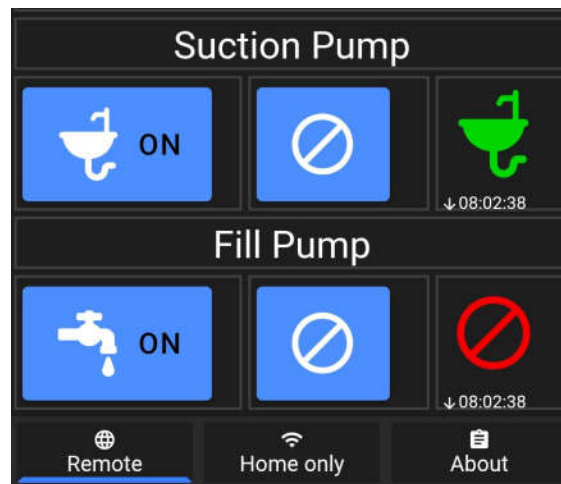
Via telegram the pump can be enabled or disabled when the **Control** button is tapped.



When the Status or Details buttons are used, the pump relay state is displayed at the bottom of each response.



On the IoT Central app, the pump relay state and controls are displayed as in the example below.

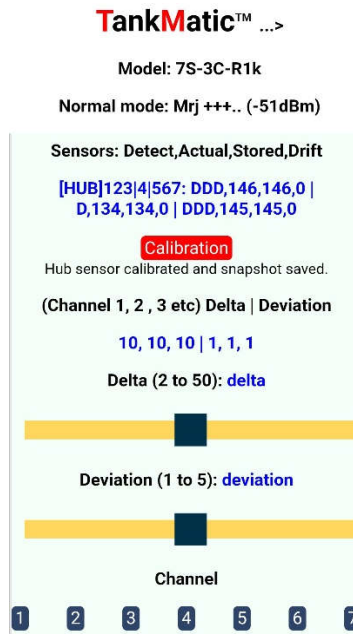


In the screenshot above the suction pump is active, the fill pump is off.

4 Calibration and General Use

4.1 Calibration

After installation, calibration of the system is recommended. If the tank (or tanks for L model) is not full, do not perform the calibration. Wait until the tank is completely full i.e. the float valve has shut off the water inside the tank.



In the detailed UI (see sub-section 3.2), in the About tab scroll to the bottom, tap the **Calibration** button, type **Yes** and tap **OK**. The Hub and Node LEDs will flash white. At the end of calibration the Visualizer will show all 7 green LEDs. The product is ready for use.

The Delta and Deviation Tweak should be left at their default values. As a general rule, the factory defaults are suitable for standard design tanks, 400 to 1000 Gallons (taller or slim profile tanks may require adjustment of the values since these have thicker walls).

- Higher values of Delta are useful for thinner walled or short tanks.
- Higher values of Deviation Tweak are for thicker walled or tall tanks. If too many Telegram notifications are being sent (i.e. water level sits exactly between two sensors), changing the value to 1 will stabilize the reading.

4.2 Use

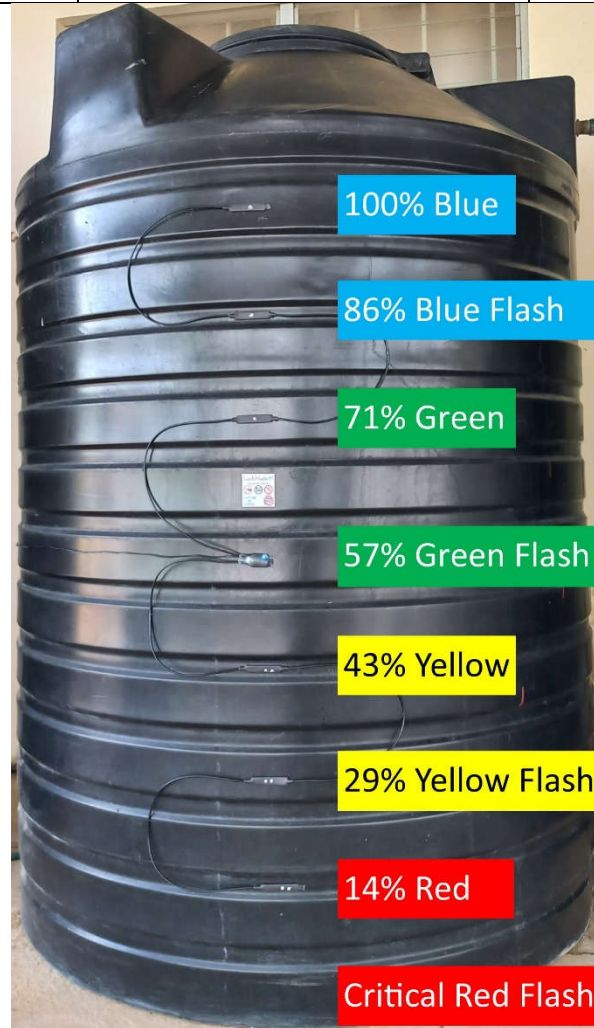
The product can be accessed:

1. Locally via its UI on your phone browser app
2. Remotely using the Telegram or IoT Central apps.

4.2.1 LED colours (units without Visualizer)

Special order products without the Visualizer ring displays information as stated below.

Level (%)	Sensor	Colour
100	S7	Blue
86	S6	Blue Flash
71	S5	Green
57	S4	Green Flash
43	S3	Yellow
29	S2	Yellow Flash
14	S1	Red
Critical	NONE	Red Flash



4.3 Troubleshooting

Please read the below to remedy issues prior to contacting SPL:

- High pressure or large flow of water onto the tank may cause a false indication of water when tank's water level is below the sensor(s). When the tank outer surface dries, the product will properly detect the internal level.

- If the product keeps going into configuration mode (cyan flashes) after successful installation, please check the WiFi router settings (some routers limit the number of connected clients). So long as the saved SSID password are correct, there is no need to re-enter the WiFi credentials; the product will reconnect when the router allows it. Typically a power cycle will solve most issues.
- If the product keeps losing connection to the internet, check that your WiFi router/extender is working properly and not releasing the product from the network: this is an issue with the client's WiFi equipment and not a fault of the product.
- If there are too many notifications being sent within a short time period, the water level may be sitting as the threshold between two sensors. Changing the Deviation Tweak to 1 may stabilize the reading.
- High current loads on the same outlet as the product (for example arc welders, heavy cutting saws etc.) may cause malfunction or damage.